

Errata

Lectures on Electromagnetic Theory

Chapter 1 — Maxwell equations and uniqueness

p. 20 complex Poynting theorem and accompanying energy definitions.

Reads:

$$\vec{\nabla} \cdot \vec{S}_c = -i2\omega \left(\frac{\mu |\vec{H}|^2}{4} - \frac{\epsilon |\vec{E}|^2}{4} \right) - \frac{\vec{E} \cdot \vec{J}^*}{2} - \frac{\vec{H}^* \cdot \vec{M}}{2}.$$

Should read:

$$\vec{\nabla} \cdot \vec{S}_c = -i2\omega \left(\frac{\mu |\vec{H}|^2}{4} - \frac{\epsilon^* |\vec{E}|^2}{4} \right) - \frac{\vec{E} \cdot \vec{J}^*}{2} - \frac{\vec{H}^* \cdot \vec{M}}{2}.$$

In the definitions following the theorem, also correct the magnetic term:

Reads: $w_m^{\text{avg}} = \frac{1}{4} \vec{H} \cdot \vec{B}^*$. *Should read:* $w_m^{\text{avg}} = \frac{1}{4} \vec{H}^* \cdot \vec{B}$.

p. 21 energy identity and its surface-integral form (and the sentence introducing them).

Reads: the difference fields are dotted "... and the second with $\delta \vec{E}^*$," giving

$$\vec{\nabla} \times \delta \vec{E} \cdot \delta \vec{H}^* - \vec{\nabla} \times \delta \vec{H} \cdot \delta \vec{E}^* = \vec{\nabla} \cdot (\delta \vec{E} \times \delta \vec{H}^*) = i\omega (\mu |\delta \vec{H}|^2 - \epsilon |\delta \vec{E}|^2),$$

$$\oint_S (\delta \vec{E} \times \delta \vec{H}^*) \cdot d\vec{s} = i\omega \int_V (\mu |\delta \vec{H}|^2 - \epsilon |\delta \vec{E}|^2) dv.$$

Should read: "... and the *conjugate of the* second with $\delta \vec{E}$," giving

$$\vec{\nabla} \times \delta \vec{E} \cdot \delta \vec{H}^* - \vec{\nabla} \times \delta \vec{H}^* \cdot \delta \vec{E} = \vec{\nabla} \cdot (\delta \vec{E} \times \delta \vec{H}^*) = -i\omega (\mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2),$$

$$\oint_S (\delta \vec{E} \times \delta \vec{H}^*) \cdot d\vec{s} = -i\omega \int_V (\mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2) dv.$$

Chapter 2 — Plane waves and waveguides

p. 33 angular-spectrum cross-reference.

Reads: "Special case where k is purely imaginary."

Should read: "Special case where k_z is purely imaginary."

p. 37 circular-polarization phase condition.

Reads:

$$|E_x| = |E_y|, \quad \delta_x - \delta_y = \frac{n\pi}{2}, \quad n \in \mathbb{Z}.$$

Should read:

$$|E_x| = |E_y| > 0, \quad \delta_x - \delta_y = \frac{(2m+1)\pi}{2}, \quad m \in \mathbb{Z}.$$

The preceding sentence should likewise require equal, nonzero amplitudes and an odd integer multiple of $\pi/2$ for the phase difference.

p. 43 phase velocity in a low-loss medium.

Reads:

$$v_p = \frac{\omega}{\beta} = \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + (\tan \delta)^2} - 1 \right] \right\}^{-1/2}.$$

Should read:

$$v_p = \frac{\omega}{\beta} = \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + (\tan \delta)^2} + 1 \right] \right\}^{-1/2}.$$

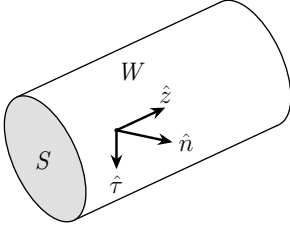
p. 53 TE-mode boundary condition at a PEC wall.

Reads: $E_\tau^0 = i\omega\mu \partial_n H_z^0$. *Should read:* $E_\tau^0 = \frac{i\omega\mu}{k_c^2} \partial_n H_z^0$.

p. 54 Figure 2.5 (illustration of a waveguide cross section): direction of the $\hat{\tau}$ arrow.

Reads: $\hat{\tau}$ is drawn pointing *up*.

Should read: $\hat{\tau}$ should point the opposite way (*downward* along the wall): $\hat{\tau} = \hat{z} \times \hat{n}$.



p. 54 propagation constant just before (2.3.13).

Reads: “If the waveguide is lossless, then $\beta = k_z$.”

Should read: “If the waveguide is lossless *and operates above cutoff*, then $\beta = k_z$.”

p. 56 two-dimensional Green identities, (2.3.19) and (2.3.21).

Reads: the middle boundary integrals use the tangential element $d\vec{l}$:

$$\oint_C g \vec{\nabla}_\perp f \cdot d\vec{l}, \quad \oint_C (g \vec{\nabla}_\perp f - f \vec{\nabla}_\perp g) \cdot d\vec{l}.$$

Should read:

$$\oint_C g \vec{\nabla}_\perp f \cdot \hat{n} dl, \quad \oint_C (g \vec{\nabla}_\perp f - f \vec{\nabla}_\perp g) \cdot \hat{n} dl.$$

p. 59 modal-orthogonality identity, Cases 1 and 2.

Reads:

$$(\hat{z} \times \vec{\nabla}_\perp \psi_{pq}) \cdot (\hat{z} \times \vec{\nabla}_\perp \psi_{p'q'}) = \hat{z} \cdot [\hat{z} (\vec{\nabla}_\perp \psi_{pq} \cdot \vec{\nabla}_\perp \psi_{pq})] = \vec{\nabla}_\perp \psi_{pq} \cdot \vec{\nabla}_\perp \psi_{pq}.$$

Should read:

$$(\hat{z} \times \vec{\nabla}_\perp \psi_{pq}) \cdot (\hat{z} \times \vec{\nabla}_\perp \psi_{p'q'}) = \hat{z} \cdot [\hat{z} (\vec{\nabla}_\perp \psi_{pq} \cdot \vec{\nabla}_\perp \psi_{p'q'})] = \vec{\nabla}_\perp \psi_{pq} \cdot \vec{\nabla}_\perp \psi_{p'q'}.$$

p. 60 triple-product identity, Case 3.

Reads: $(\hat{z} \times \vec{\nabla}_{\perp} \psi_{pq}) \cdot \hat{n} = \vec{\nabla}_{\perp} \psi_{pq} \cdot (\hat{z} \times \hat{n}) = -\vec{\nabla}_{\perp} \psi_{pq} \cdot \hat{\tau}.$

Should read: $(\hat{z} \times \vec{\nabla}_{\perp} \psi_{pq}) \cdot \hat{n} = \vec{\nabla}_{\perp} \psi_{pq} \cdot (\hat{n} \times \hat{z}) = -\vec{\nabla}_{\perp} \psi_{pq} \cdot \hat{\tau}.$

Chapter 3 — Radiation

p. 76 Hertzian dipole, the intermediate $\vec{\nabla}[\vec{\nabla} \cdot (\hat{z}G)]$.

Reads:

$$\vec{\nabla}[\vec{\nabla} \cdot (\hat{z}G)] = G \left[\left(-k^2 + \frac{2ik}{r} + \frac{1}{r^2} \right) \cos \theta \hat{r} + \left(ik + \frac{1}{r} \right) \sin \theta \hat{\theta} \right].$$

Should read:

$$\vec{\nabla}[\vec{\nabla} \cdot (\hat{z}G)] = G \left[\left(-k^2 + \frac{2ik}{r} + \frac{2}{r^2} \right) \cos \theta \hat{r} + \frac{1}{r} \left(ik + \frac{1}{r} \right) \sin \theta \hat{\theta} \right].$$

p. 82 rectangular-aperture far field, phase factor and integration limits.

Reads:

$$\int_{-a/2}^{a/2} \int_{-b/2}^{b/2} (\dots) e^{ik(x'u+y'v)} dx' dy'.$$

Should read:

$$\int_{-b/2}^{b/2} \int_{-a/2}^{a/2} (\dots) e^{ik(x'u+y'v)} dx' dy'.$$

Here a is the aperture width along x and b along y . Keeping the order $dx' dy'$ requires the inner limits to be $\pm a/2$ and the outer limits to be $\pm b/2$.

p. 82 rectangular-aperture far field, amplitude prefactor.

Reads: $\vec{E} \approx ikE_0ab \frac{e^{-ikr}}{4\pi r} (\dots) \text{sinc}\left(\frac{ka}{2}u\right) \text{sinc}\left(\frac{kb}{2}v\right).$

Should read: $\vec{E} \approx ikE_0ab \frac{e^{-ikr}}{2\pi r} (\dots) \text{sinc}\left(\frac{ka}{2}u\right) \text{sinc}\left(\frac{kb}{2}v\right).$

Chapter 4 — Electromagnetic scattering

p. 91 cylindrical-wave (Jacobi–Anger) expansion, projection step.

Reads: “Multiplying [the expansion] by $e^{im\phi}$ and integrating over $\phi \dots$ ”

Should read: “Multiplying [the expansion] by $e^{-im\phi}$ and integrating over $\phi \dots$ ”

p. 96 separation of variables, scalar Helmholtz equation.

Reads: “... multiply $r^2 \sin \theta$ and divide it by $\psi \dots$ ” *Should read:* “... multiply $r^2 \sin^2 \theta$ and divide it by $\psi \dots$ ”

p. 97 Legendre regularity condition.

Reads: “To ensure that **all solutions** of the equation remain finite at **$\sin\theta = \pm 1$** , the parameter μ^2 must take the specific form $\mu^2 = n(n+1)$, where n is a non-negative integer.”

Should read: “For **a nonzero solution** to remain finite at **both poles**, **$\cos\theta = \pm 1$** , the parameter must be $\mu^2 = n(n+1)$, where n is a non-negative integer. **The regular solution is proportional to $P_n(\cos\theta)$.**”

p. 100 incident-field Debye-potential expansion.

Reads: $\pi_e^i = \sum_{n=0}^{\infty} \sum_{m=0}^n j_n(k_2 r) P_n^m(\cos\theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi)$.

Should read: $\pi_e^i = \sum_{n=0}^{\infty} \sum_{m=0}^n j_n(kr) P_n^m(\cos\theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi)$.

p. 101 Riccati–Bessel identity.

Reads: $\frac{d^2}{dr^2} [r j_n(kr)] + (kr)^2 j_n(kr) = n(n+1) j_n(kr)$.

Should read: $\left(\frac{d^2}{dr^2} + k^2 \right) [r j_n(kr)] = \frac{n(n+1)}{r} j_n(kr)$.

p. 98 and 104 second-kind spherical Hankel function (two occurrences).

Reads:

$$h_n^{(2)}(kr) = \sqrt{\frac{\pi}{2kr}} H_{n+1/2}(kr), \quad \hat{h}_n^{(2)}(kr) = \sqrt{\frac{\pi kr}{2}} H_{n+1/2}(kr) \approx i^{n+1} e^{-ikr}.$$

Should read:

$$h_n^{(2)}(kr) = \sqrt{\frac{\pi}{2kr}} H_{n+1/2}^{(2)}(kr), \quad \hat{h}_n^{(2)}(kr) = \sqrt{\frac{\pi kr}{2}} H_{n+1/2}^{(2)}(kr) \approx i^{n+1} e^{-ikr}.$$

p. 102 dielectric-sphere matching conditions (ϵ and μ interchanged in the algebraic continuity of the Debye potentials, and one undefined subscript).

Reads:

$$\mu \pi_e = \mu_d \pi_e^d, \quad \epsilon \pi_{mr} = \epsilon_d \pi_m^d.$$

Should read:

$$\epsilon \pi_e = \epsilon_d \pi_e^d, \quad \mu \pi_m = \mu_d \pi_m^d.$$

p. 103 dielectric-sphere internal Mie coefficients c_n, d_n (numerators).

Here D_e and D_m abbreviate the unchanged denominators printed in the book:

$$D_e = \sqrt{\epsilon\mu_d} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d\mu} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a),$$

$$D_m = \sqrt{\epsilon_d\mu} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon\mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a).$$

Reads:

$$c_n = \frac{i\sqrt{\epsilon_d\mu}}{D_e}, \quad d_n = \frac{i\sqrt{\epsilon\mu_d}}{D_m},$$

Should read:

$$c_n = \frac{i\mu_d\sqrt{\epsilon_d/\mu}}{D_e}, \quad d_n = \frac{i\mu_d\sqrt{\epsilon_d/\mu}}{D_m},$$

p. 104 far-scattered electric field, (4.5.31) and (4.5.32).

Reads:

$$E_\theta \approx f_\theta(\theta, \phi) \frac{e^{-ikr}}{r}, \quad E_\phi \approx f_\phi(\theta, \phi) \frac{e^{-ikr}}{r}.$$

Should read:

$$E_\theta \approx \textcolor{red}{E}_0 f_\theta(\theta, \phi) \frac{e^{-ikr}}{r}, \quad E_\phi \approx \textcolor{red}{E}_0 f_\phi(\theta, \phi) \frac{e^{-ikr}}{r}.$$

The definitions of f_θ , f_ϕ , and the cross sections are unchanged.

p. 105 the $|S_1(\theta)|^2$ and $|S_2(\theta)|^2$ expansions (bodies interchanged, and one mis-paired term).

With $c_n = (2n+1)/[n(n+1)]$:

Reads:

$$\begin{aligned} |S_1|^2 &= \sum_{n,n'} c_n c_{n'} [a_n \textcolor{red}{\tau}_n a_{n'}^* \textcolor{red}{\tau}_{n'} + b_n \pi_n b_{n'}^* \pi_{n'} + a_n \textcolor{red}{\tau}_n b_{n'}^* \pi_{n'} + b_n \pi_n a_{n'}^* \pi_{n'}], \\ |S_2|^2 &= \sum_{n,n'} c_n c_{n'} [a_n \pi_n a_{n'}^* \pi_{n'} + b_n \textcolor{red}{\tau}_n b_{n'}^* \textcolor{red}{\tau}_{n'} + a_n \pi_n b_{n'}^* \textcolor{red}{\tau}_{n'} + b_n \textcolor{red}{\tau}_n a_{n'}^* \pi_{n'}]. \end{aligned}$$

Should read:

$$\begin{aligned} |S_1|^2 &= \sum_{n,n'} c_n c_{n'} [a_n \pi_n a_{n'}^* \pi_{n'} + b_n \textcolor{red}{\tau}_n b_{n'}^* \textcolor{red}{\tau}_{n'} + a_n \pi_n b_{n'}^* \textcolor{red}{\tau}_{n'} + b_n \textcolor{red}{\tau}_n a_{n'}^* \pi_{n'}], \\ |S_2|^2 &= \sum_{n,n'} c_n c_{n'} [a_n \textcolor{red}{\tau}_n a_{n'}^* \textcolor{red}{\tau}_{n'} + b_n \pi_n b_{n'}^* \pi_{n'} + a_n \textcolor{red}{\tau}_n b_{n'}^* \pi_{n'} + b_n \pi_n a_{n'}^* \textcolor{red}{\tau}_{n'}]. \end{aligned}$$

p. 105 π/τ orthogonality normalization.

Reads: $\int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta d\theta = \frac{2n(n+1)^2}{2n+1} \quad (n = n').$

Should read: $\int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta d\theta = \frac{2n^2(n+1)^2}{2n+1} \quad (n = n').$

p. 106 Rayleigh (small-sphere) scattering coefficients, overall sign.

Reads:

$$a_1 = -\frac{2i}{3}(ka)^3 \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon}, \quad b_1 = -\frac{2i}{3}(ka)^3 \frac{\mu_d - \mu}{\mu_d + 2\mu}.$$

Should read:

$$a_1 = +\frac{2i}{3}(ka)^3 \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon}, \quad b_1 = +\frac{2i}{3}(ka)^3 \frac{\mu_d - \mu}{\mu_d + 2\mu}.$$

p. 106 PEC-sphere Mie coefficients.

Reads:

$$a_n = -\frac{\hat{j}'_n(ka)}{\hat{h}_{n(2)}'(ka)}, \quad b_n = -\frac{\hat{j}_n(ka)}{\hat{h}_{n(2)}(ka)}.$$

Should read:

$$a_n = +\frac{\hat{j}'_n(ka)}{\hat{h}_{n(2)}'(ka)}, \quad b_n = +\frac{\hat{j}_n(ka)}{\hat{h}_{n(2)}(ka)}.$$

Chapter 5 — Special relativity

p. 111 derivation of the Lorentz transformation.

Reads: “Since by **(P2)** it is required that the transformation is linear ...”

Should read: “By **(P1)**, together with homogeneity of space and time and coincident origins, we use a linear transformation ...”

p. 117 skew-tensor square identity.

Reads:

$$(\vec{\beta})^2 \cdot \vec{u} = \vec{\beta} \times (\vec{\beta} \times \vec{u}) = \vec{\beta} \vec{\beta} \cdot \vec{u} - \beta^2 \vec{u}.$$

Should read:

$$\vec{\beta}^2 \cdot \vec{u} = \vec{\beta} \cdot (\vec{\beta} \cdot \vec{u}) = \vec{\beta} \times (\vec{\beta} \times \vec{u}) = \vec{\beta} \vec{\beta} \cdot \vec{u} - \beta^2 \vec{u}.$$

p. 118 identity used in deriving $c\vec{B}'$.

Reads:

$$\begin{aligned} \vec{\beta} \cdot [(\vec{\alpha} \cdot \vec{\nabla}') \times \vec{E}] &= \vec{\beta} \cdot \left[\left(\vec{\nabla}' + (\gamma - 1) \frac{\vec{\beta} \cdot \vec{\nabla}'}{\beta^2} \vec{\beta} \right) \times \vec{E} \right] \\ &= \vec{\beta} \cdot (\vec{\nabla}' \times \vec{E}) = -\vec{\nabla}' \cdot (\vec{\beta} \times \vec{E}). \end{aligned}$$

Should read:

$$\begin{aligned} \vec{\beta} \cdot [(\vec{\alpha} \cdot \vec{\nabla}') \times \vec{E}] &= \vec{\beta} \cdot \left[\left(\vec{\nabla}' + (\gamma - 1) \frac{\vec{\beta} \cdot \vec{\nabla}'}{\beta^2} \vec{\beta} \right) \times \vec{E} \right] \\ &= \vec{\beta} \cdot (\vec{\nabla}' \times \vec{E}) = -\vec{\nabla}' \cdot (\vec{\beta} \times \vec{E}). \end{aligned}$$

p. 119 intermediate divergence and curl identities before (5.2.15).

In the first displayed equation, insert the divergence dot:

Reads: $\gamma \vec{\beta} (\vec{\alpha} \cdot \vec{\nabla}') c\vec{B}$. *Should read:* $\gamma \vec{\beta} (\vec{\alpha} \cdot \vec{\nabla}') \cdot c\vec{B}$.

In the following identity:

Reads:

$$\vec{\beta} [(\vec{\alpha} \cdot \vec{\nabla}') c\vec{B}] - \vec{\alpha} \cdot (\vec{\beta} \cdot \vec{\nabla}') c\vec{B} = \vec{\beta} (\vec{\nabla}' \cdot c\vec{B}) - (\vec{\beta} \times \vec{\nabla}') c\vec{B}.$$

Should read:

$$\begin{aligned} \vec{\beta} [(\vec{\alpha} \cdot \vec{\nabla}') \cdot c\vec{B}] - \vec{\alpha} \cdot (\vec{\beta} \cdot \vec{\nabla}') c\vec{B} \\ = \vec{\beta} (\vec{\nabla}' \cdot c\vec{B}) - (\vec{\beta} \cdot \vec{\nabla}') c\vec{B} = \vec{\nabla}' \times (\vec{\beta} \times c\vec{B}). \end{aligned}$$

Appendix B — Bessel and Hankel functions

p. 133 physical roles of the Hankel functions.

Reads: “... representing *outward and inward* cylindrical waves, respectively.”

Should read: “... representing *inward and outward* cylindrical waves, respectively.”

p. 134 finite-interval integral representation.

Reads:

$$J_\nu(x) = \frac{1}{\pi} \int_0^\pi \cos(\nu\theta - x \sin \theta) d\theta.$$

Should read:

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta, \quad n \in \mathbb{Z}.$$

Appendix C — Legendre functions

p. 138 Legendre polynomial $P_5(x)$, missing factor of x .

Reads: $P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15)$. *Should read:* $P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$.

p. 139 Figure C.1 caption.

Reads: “Legendre polynomials for $n = 0, 1, 2, 3, 4, 5$.”

Should read: “Legendre polynomials for $n = 0, 1, 2, 3, 4$.”

p. 139 associated-Legendre convention, (C.3.1).

Reads:

$$P_n^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad m = 0, 1, \dots, n.$$

Should read:

$$P_n^m(x) = (1-x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad m = 0, 1, \dots, n.$$

This is the phase-free convention used in Chapter 4. The Condon–Shortley convention includes an additional factor $(-1)^m$: $P_{n,\text{CS}}^m = (-1)^m P_n^m$.

p. 140 Legendre function of the second kind, closed form.

Reads:

$$Q_n(x) = \frac{1}{2} P_n(x) \ln\left(\frac{1+x}{1-x}\right) - \sum_{k=1}^n \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots (2k)} P_{n-k}(x).$$

Should read:

$$Q_n(x) = \frac{1}{2} P_n(x) \ln\left(\frac{1+x}{1-x}\right) - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x).$$